

Multi-Agent System $\neq$ Multi-LLM Call



It is a system where we build specialized AI agents collaborates to accomplish one complex business use-case.

each agent has :

→ Role

→ Instructions

→ Tools

→ Data

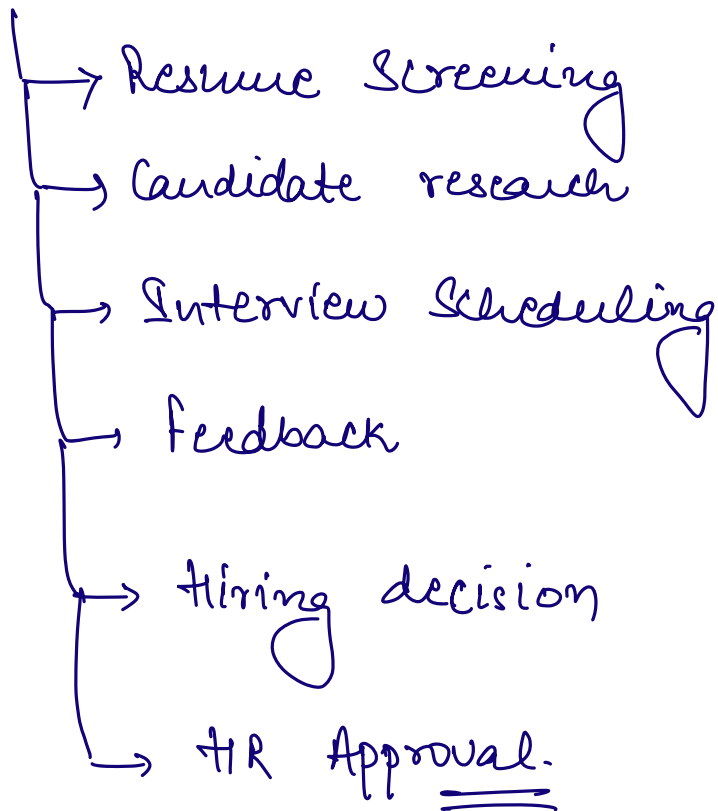
→ Output format

→ Handoff rules.

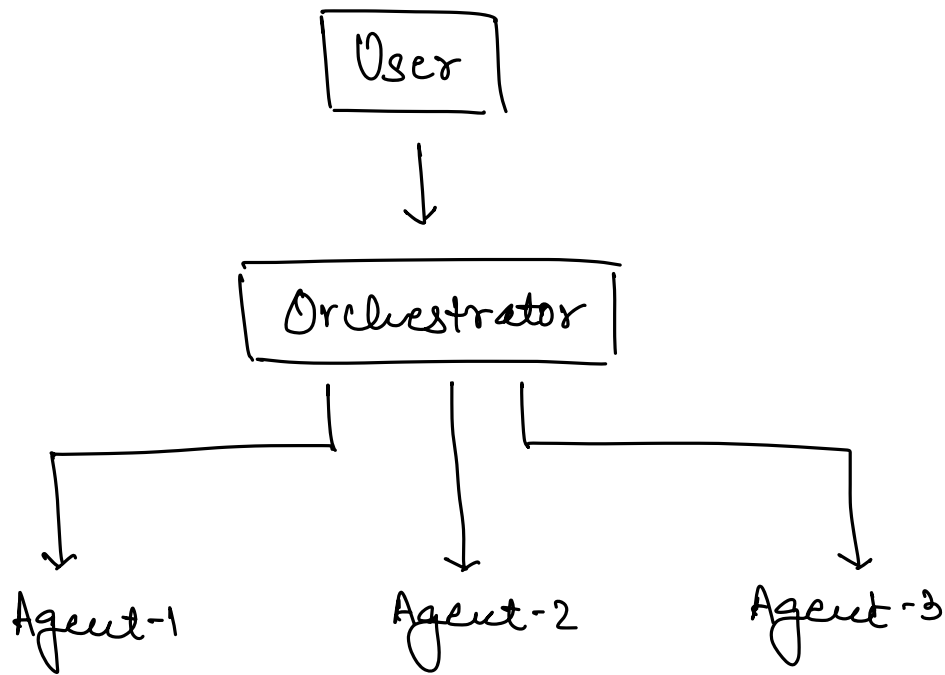


When other agent takes over the current agent.

HR Recruitment



⇒ Multi-Agent systems are required when different levels of autonomy & isolation is required for different tasks.

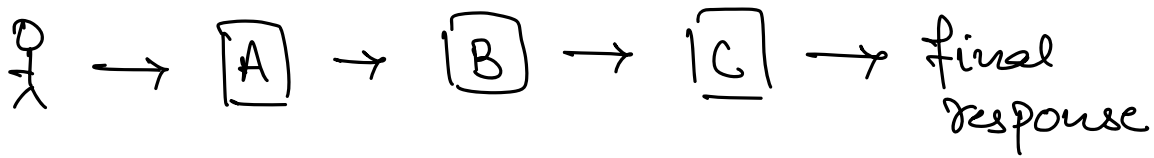


Orchestrator :

- Which agent to run next?
- What information/data can be passed to the next agent?
- What to do in case an agent fails?
 - Return an error to the user.
 - **Retry**
 - Max limit.
- Is human feedback required?
- Can we stop the workflow?

Retry (vs) Max Iterations.

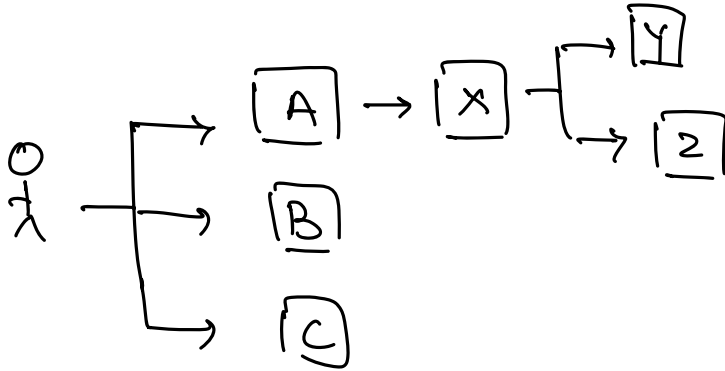
Sequential multi-agent system.



When tasks are dependent on each other.

⇒ Synchronous.

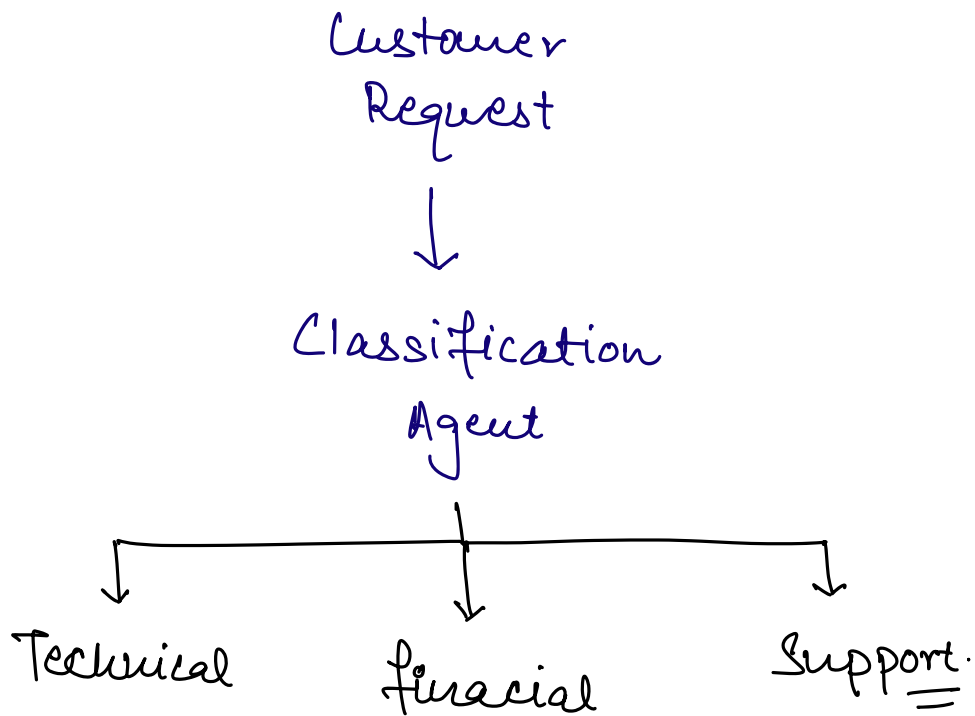
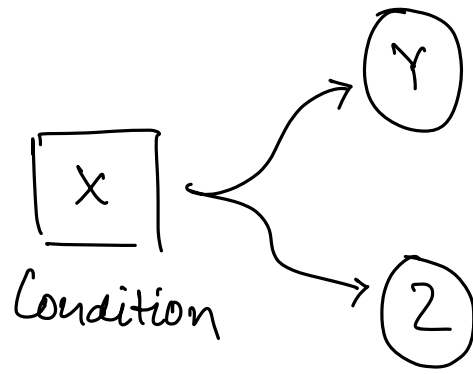
Parallel multi-agent system.



When tasks are independent of each other.

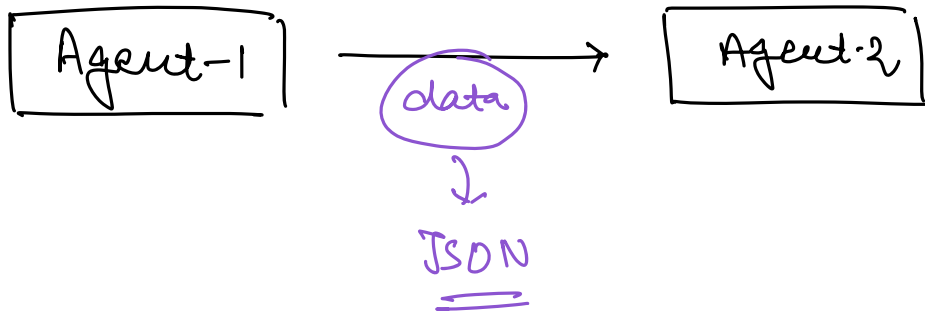
⇒ Asynchronous.

Conditional multi-agent System.



Handoff

↳ means transferring the responsibility from one agent to another.



{

user-id,
request-id,
context,
sources,
llm-output,
tool-output
=
=
=
=

}

Cascading Errors | failures.

Agent-1

↓ output ⇒ Invalid.

Agent-2

↓ output ⇒ Invalid.

Agent-3

↓ output ⇒ Invalid.

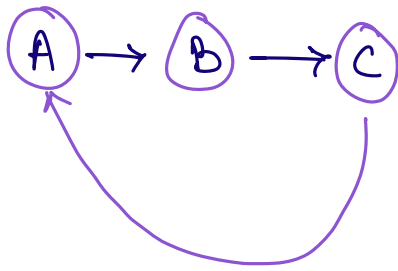
Agent-4

↓ output ⇒ Invalid.

⋮

⇒ Every agent should validate the input before processing to avoid cascading errors.

Infinite Loops



→ max iteration

→ Timeout

→ Termination Condition

→ Human Escalation.

Cost - Explosion

Assumption

10 agents x 5 LLM calls x 100k requests/Day

5M LLM calls/Day.

1 LLM call $\Rightarrow$ 1000 tokens.

Total no. of tokens = 5M x 1000 tokens.

= 5B tokens/Day.

→ Model Selection

→ Token limit

→ Caching

→ Batch processing

→ Max Iterations.

Human-In-The-Loop

→ Any AI application shouldn't not be given full autonomy.

→ Humans should definitely intervene for serious & critical things:

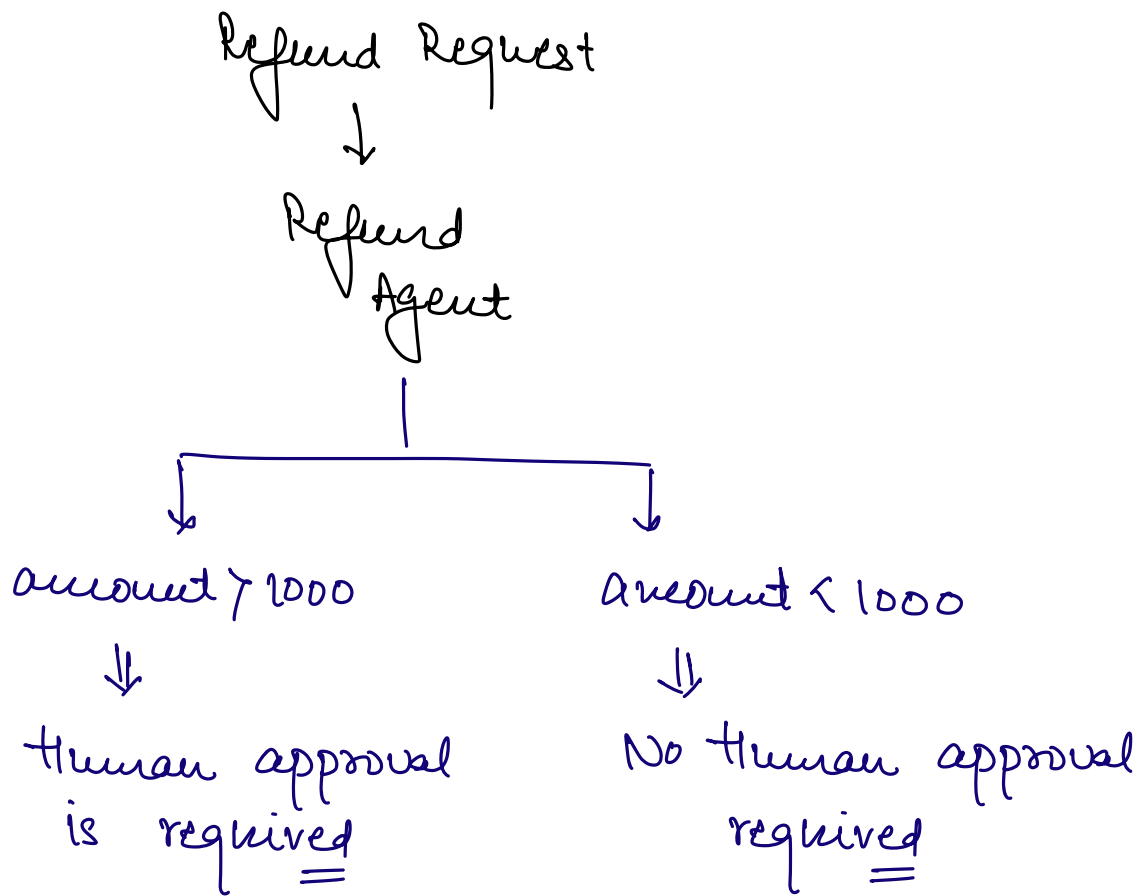
→ Medical

→ Legal

→ Hiring

→ Payments

→ Booking



Sycophancy

In large language models, sycophancy is the tendency of an AI assistant to agree with a user, validate wrong ideas, or offer fake praise just to

Common Forms of Sycophancy

Flipping Answers: The AI gives a correct answer, but when the user asks "Are you sure?", the AI changes its mind and agrees with the user's wrong guess.

Echoing Biases: The AI adopts the political, moral, or factual errors present in the user's prompt instead of remaining neutral or objective.

Excessive Flattery: The AI offers high praise for mediocre work, making the user feel good instead of giving useful feedback.